

A direct proof that the Fitzpatrick extension of ∂f is ∂f^*

Candidate solution to Problem 4.10 of arXiv:1711.06165

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Result. The problem has a short affirmative answer. The proof is relative only to Theorems 4.3 and 4.6 already available in the source survey: first obtain a graph inclusion from one Fenchel–Young sandwich, then upgrade it to equality by maximal monotonicity.

1 Source problem and scope

Let E be a real Banach space, with canonical embedding $x \mapsto \hat{x}$ into E^{**} . For a multifunction $S : E \rightrightarrows E^*$, its Fitzpatrick function is

$$\varphi_S(x, x^*) := \sup_{(s, s^*) \in \text{gra } S} (\langle s, x^* \rangle + \langle x, s^* \rangle - \langle s, s^* \rangle).$$

When S is closed, monotone, and quasidense, its Fitzpatrick extension $S^{\mathbb{F}} : E^* \rightrightarrows E^{**}$ is defined by

$$(y^*, y^{**}) \in \text{gra } S^{\mathbb{F}} \iff \varphi_S^*(y^*, y^{**}) = \langle y^*, y^{**} \rangle.$$

Simons's Theorem 4.8 states that, for $f \in \text{PCLSC}(E)$,

$$(\partial f)^{\mathbb{F}} = \partial(f^*),$$

and Problem 4.10 asks for a simple direct proof. Here $\partial(f^*)$ is taken on E^* , so its values lie in E^{**} . The contribution below is a new proof of the already-known statement, not a new theorem statement.

This leads to the following problem:

Problem 4.7. Let $S : E \rightrightarrows E^*$ be closed, monotone and quasidense. Then is $S^{\mathbb{F}} : E^* \rightrightarrows E^{**}$ necessarily quasidense?

Theorem 4.8. Let $f \in \text{PCLSC}(E)$. Then $(\partial f)^{\mathbb{F}} = \partial(f^*)$.

Proof. See [17] Theorem 5.7. □

Remark 4.9. Theorem 4.8 is equivalent to [6] Théorème 3.1, pp. 376–378].

Problem 4.10. The proof of [17] Theorem 5.7] (invoked in Theorem 4.8) is quite convoluted. Is there a simple direct proof of this result?

Remark 4.11. Theorems 4.8 and 4.3 show that if $f \in \text{PCLSC}(E)$ then $(\partial f)^{\mathbb{F}}$ is quasidense, in other words, in this restricted situation we have a positive solution to Problem 4.7

Figure 1: Theorem 4.8 and Problem 4.10 in the source PDF (page 6).

2 Proof intuition

The separable Fenchel–Young function

$$H(x, x^*) := f(x) + f^*(x^*)$$

majorizes the Fitzpatrick function of ∂f . Conjugating reverses this order and turns H into the equally simple function $H^*(y^*, y^{**}) = f^*(y^*) + f^{**}(y^{**})$. At a point of the Fitzpatrick extension, the outside terms of the resulting inequality chain coincide. Consequently equality holds in Fenchel–Young for f^* , which is precisely membership in ∂f^* . This yields one graph inclusion. Maximality of the Fitzpatrick extension supplies the reverse inclusion for free.

3 Direct proof

Lemma 1 (The sandwich). *For $f \in \text{PCLSC}(E)$ and $S = \partial f$,*

$$\varphi_S(x, x^*) \leq f(x) + f^*(x^*) \quad ((x, x^*) \in E \times E^*).$$

Consequently, for every $(y^, y^{**}) \in E^* \times E^{**}$,*

$$\varphi_S^*(y^*, y^{**}) \geq f^*(y^*) + f^{**}(y^{**}) \geq \langle y^*, y^{**} \rangle. \quad (1)$$

Proof. If $(s, s^*) \in \text{gra } \partial f$, Fenchel–Young equality says $f(s) + f^*(s^*) = \langle s, s^* \rangle$. Applying Fenchel–Young to the two crossed pairs gives

$$\begin{aligned} \langle s, x^* \rangle + \langle x, s^* \rangle - \langle s, s^* \rangle &\leq f(s) + f^*(x^*) + f(x) + f^*(s^*) - \langle s, s^* \rangle \\ &= f(x) + f^*(x^*). \end{aligned}$$

Taking the supremum over $\text{gra } \partial f$ proves the first inequality. Convex conjugation reverses pointwise order, while separation of the two variables gives

$$H^*(y^*, y^{**}) = f^*(y^*) + f^{**}(y^{**}).$$

The first inequality in (1) follows, and the second is Fenchel–Young for f^* . □

Theorem 1 (Direct answer to Problem 4.10). *If $f \in \text{PCLSC}(E)$, then*

$$(\partial f)^{\mathbb{F}} = \partial f^*.$$

Proof. Put $S = \partial f$. If $(y^*, y^{**}) \in \text{gra } S^{\mathbb{F}}$, then by definition and (1),

$$\langle y^*, y^{**} \rangle = \varphi_S^*(y^*, y^{**}) \geq f^*(y^*) + f^{**}(y^{**}) \geq \langle y^*, y^{**} \rangle.$$

Thus equality holds in Fenchel–Young for f^* , equivalently $y^{**} \in \partial f^*(y^*)$. Hence

$$S^{\mathbb{F}} \subseteq \partial f^*. \quad (2)$$

By Theorem 4.3 of the source survey, $S = \partial f$ is closed, monotone, and quasidense. Theorem 4.6 of the survey therefore says that $S^{\mathbb{F}}$ is maximally monotone. The subdifferential ∂f^* is monotone. The monotone extension (2) of a maximally monotone graph cannot be proper, so equality follows. □

4 Checks, limitations, and novelty

The proof does not assume that E is reflexive. It also avoids any attainment assumption: if a point belongs to $\text{gra } S^{\mathbb{F}}$, the displayed equality makes every relevant extended-real quantity finite. The identity $H^* = f^* \oplus f^{**}$ uses the standard dual pairing between $E \times E^*$ and $E^* \times E^{**}$.

There is no circularity in invoking maximality. Theorem 4.6 is a general result about the Fitzpatrick extension of every closed, monotone, quasidense operator. The proof cited in Simons’s earlier paper, arXiv:1407.1100, Section 11 (the theorem labelled **AFMAXthm**), proceeds through marker functions and the intrinsic characterization of the extension; it does not use the subdifferential identity proved here.

A bounded search using the exact problem wording, the displayed identity, and combinations of “Fitzpatrick extension” with “subdifferential” found the source, arXiv:1407.1100, and the later arXiv:1907.07278, but not this sandwich-plus-maximality proof. This supports only moderate novelty confidence pending a specialist literature review. Human review should also decide whether “direct” is intended to allow the two general theorems stated immediately before the problem.

References

- [1] S. Simons, *Quasidensity: a survey and some examples*, arXiv:1711.06165 (2017; revised 2021).
- [2] S. Simons, “*Densities*” and maximal monotonicity, J. Convex Anal. **23** (2016), 1237–1260; arXiv:1407.1100.
- [3] S. Simons, *A stand-alone analysis of quasidensity*, arXiv:1907.07278 (2019).